

The effects of medication education and behavioral intervention on Chinese patients with epilepsy



Fengmin Tang^{a,b}, Guoxing Zhu^{c,*}, Zheng Jiao^a, Chunlai Ma^a, Nianzu Chen^b, Bin Wang^{a,*}

^a Department of Pharmacy, Hospital of Huashan, Fudan University, Shanghai, China

^b Department of Pharmacy, Hospital of Eye, Ear, Nose and Throat, Fudan University, Shanghai, China

^c Department of Neurology, Hospital of Huashan, Fudan University, Shanghai, China

ARTICLE INFO

Article history:

Received 7 March 2014

Revised 19 May 2014

Accepted 23 May 2014

Available online xxxx

Keywords:

Adherence

Medication education

Behavioral intervention

Epilepsy

China

Pharmacist

Four-item Morisky Medication Adherence Scale

ABSTRACT

Objectives: The objectives of this study were to evaluate the effects of medication education and behavioral intervention on Chinese patients with epilepsy and to compare the difference between them.

Methods: A total of 109 patients with epilepsy who did not take their antiepileptic drugs (AEDs) more than once were randomly assigned to two intervention groups: the medication education group (group I) and the medication education with behavioral intervention group (group II). Group I was initially provided with medication education in the form of oral education and written materials, and this education was reinforced by monthly calls from the pharmacist over the next six months. The behavioral intervention provided to group II consisted of a modified medication schedule which was based on cue-dose training therapy. The outcomes that were evaluated both in the beginning and in the end of the study included adherence, which was measured using the four-item Morisky Medication Adherence Scale (MMAS-4), the number of seizures, knowledge of AEDs, and the number of patients who missed a dose of their AEDs. Differences within and between the groups were analyzed. **Results:** After intervention, the adherence and knowledge of AEDs increased greatly in all patients, and the number of patients who had seizures or missed AEDs decreased. However, no significant differences were observed between groups I and II. The observed changes were (group I vs group II, p value) increased adherence: 62.3% vs 64.3%, 0.827; increased knowledge of AEDs: 88.7% vs 80.4%, 0.231; and improved seizure control: 64.2% vs 64.3%, 0.988. In addition, the percentage of patients who forgot to take their AEDs decreased to 45.0% from more than 70%, and 44.9% of these patients took the missed AEDs as soon as they remembered.

Discussion: These findings clearly demonstrate that medication education and reinforced telephone calls from pharmacists can help to increase adherence to AEDs, the knowledge of patients regarding AEDs, and seizure control. However, the inclusion of a behavioral strategy that was easy to administer and use in this program did not lead to any significant effects on improving adherence. The results indicate that pharmacists can play an important role in improving the effects of medication regimens, but further research is required to identify strategies for improving adherence to behavioral theory.

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1. Introduction

Poor adherence to treatment regimens for chronic diseases is an increasing worldwide problem and is one of the primary causes of treatment failure in drug therapy. Data suggest that patients exhibit an approximately 50% adherence to long-term therapy for chronic illnesses in developed countries, and the rates are even lower in developing countries [1]. This poor adherence increases the burden of chronic disease worldwide. Epilepsy is a chronic disease for which antiepileptic

drugs (AEDs) need to be taken for a long period of time. Adherence to AEDs is also suboptimal, and the rate of nonadherence ranges from 30% to 50% [2]. Nonadherence to AEDs is associated with higher rates of serious clinical events, including in a threefold increase in mortality risk [3] and an increased likelihood and cost of hospitalization and emergency room treatment [4]. Most people who are afflicted with epilepsy live in developing countries, with over half of the world's 50 million patients with epilepsy living in Asia [5]. There are nine million cases in China alone [6], which accounts for nearly one-sixth of the cases in the world. Therefore, nonadherence to AEDs is a large problem in China that needs to be solved.

The World Health Organization (WHO) has concluded that "increasing the effectiveness of adherence interventions may have a far greater impact on the health of the population than any improvement in specific medical treatments" [1]. The effects of various

* Corresponding authors at: Department of Pharmacy, Huashan Hospital, No. 12 of Middle Wu Lu Mu Qi Road, Shanghai 200040, China. Tel.: +86 21 52888380; fax: +86 21 62482068.

E-mail address: tangfengm2008@163.com (F. Tang).

interventions to improve adherence have been proven in many studies [7–10]. Many researchers consider educational and behavioral strategies to be the two main interventions that can lead to improved adherence [11,12]. Although education has been shown to have mixed effects on adherence, behavioral interventions have been suggested to have better effects on adherence [12]. Providing instruction about the methods of incorporating drug administration into patients' daily lives, as one type of behavioral intervention, was also found to improve adherence to AEDs; however, no studies demonstrated this [1]. In all of the current studies regarding improving medication adherence for chronic health problems, researchers have compared a single treatment to "usual care", in which no specific intervention is performed; no published studies have compared two different interventions [12]. In this study, we not only evaluated the effects of medication education and behavioral intervention on patients with epilepsy but also compared the effects of the two intervention strategies to reveal the value of behavioral intervention. Based on the prominence of Fudan University's Huashan Hospital in treating epilepsy, patients with epilepsy were the first targeted population we evaluated to explore the effect of interventions on improved adherence in this program.

2. Methods

2.1. Participants

We focused on patients with epilepsy who were treated between September 2011 and March 2013 at the outpatient clinic of Neurology in Huashan Hospital, University of Fudan, Shanghai, China. Eligible patients were recruited if they were diagnosed with epilepsy, were older than 16 years of age, took antiepileptic drugs for more than six months, and did not take their AEDs at least once over the past six months. Patients were excluded if they only took traditional Chinese medicine (TCM)/Chinese patent medicine, had barriers in communication or cognitive impairment and could not communicate with researchers, stopped taking antiepileptic drugs, or lost contact immediately after being recruited into the study. The attending physician asked each patient whether he or she had forgotten to take their AEDs at any point within the past six months.

All patients who met the inclusion criteria in the clinic of Doctor Guoxing Zhu were included in the study and assigned to one of the two groups with different levels of intervention. One group was provided medication education (group I), and the other group was provided medication education and behavioral intervention (group II). A list of 300 random numbers between "0" and "9" was generated using IBM SPSS statistics 19. The patients were numbered according to the order in which they were recruited. Patients who had received an even randomly generated number were assigned to group I, and patients who received odd numbers were assigned to group II.

The Research Ethics Committee of Huashan Hospital approved this study. Among the 154 eligible patients who were approached for informed consent, 124 initially volunteered to participate in the study.

2.2. Interventions

2.2.1. Educational intervention

2.2.1.1. Oral and written education. All patients were educated by a pharmacist according to the guidelines of the American Society of Health-System Pharmacists (ASHP) regarding pharmacist-conducted patient education and counseling [13]. After their baseline knowledge of AEDs was assessed, all patients were educated orally. Written material was then given to the patients to provide them with more details about AEDs.

2.2.1.2. Monthly calls from the pharmacist. To strengthen the effects of medication education, the patients were followed up by telephone every month by the pharmacist to check whether they were nonadherent. Patients were again instructed about their medications and asked to adhere to their AEDs.

2.2.2. Behavioral intervention

The behavioral intervention utilized in this program consisted of a modified medication schedule, which was based on cue–dose training therapy [14] and the strategies described by Roter et al. [11]. The design of the modified medication schedule was modeled after the schedules developed by Kripalani et al. [15] and Cordasco et al. [8].

The medication schedule used in this program was presented in the form of a table that illustrated the daily medication therapy of patients with pictures of AEDs, and it provided patients with cues to take their medications (see Fig. 1). It was modified from a useful tool reported by Cordasco with permission. This individualized schedule was developed for each patient, and the pharmacist pasted a picture of the AEDs that each patient was taking in the first column along with a picture corresponding to the number of pills in the column for "Breakfast," "Lunch," "Dinner," or "Bedtime." In the last column, the pharmacist wrote down the most common activities that patients were performing each time they took their medication and the approximate time they would be taking it. This last column was included to help patients incorporate AED administration into their daily lives and to transform their daily routines into cues to take their AEDs. This method is based on the hypothesis that cue–dose training, which uses personalized cues for remembering particular dose times, can improve medication adherence [14]. It has been reported that the most common cue for patients was to link the dose with an event in their daily life [16,17], such as embedding medication habits at mealtime or during the patients' standard wake-up and sleep routines. Coincidentally, the most common cue for the patients involved in the current program was mealtime.

2.3. Measures

2.3.1. Adherence

Self-report methods have been widely used to study adherence in epilepsy [18]. In this program, adherence was measured using the four-item Morisky Medication Adherence Scale (MMAS-4) [19], which is a widely used and validated self-report method. This scale is categorized into three levels: low, medium, and high. The MMAS-4 is a feasible, reliable, and valid measure of patient adherence that can rapidly assess the adherence of patients at a low cost [20]. In addition, it is easy to use in clinical settings and has been reported to be comparable with pill counts, electronic monitoring, drug refills [21,22], and other measures of adherence [23]. The MMAS-4 is considered to be the correct measure to evaluate adherence in clinical trials. Furthermore, the MMAS-4 has been shown to be adaptable to in China, and its validity has been proven by many researchers in China [24–26].

Each item of the four items was answered by a simple "yes" or "no". A response of "yes" was scored as 0 point, and a "no" response was counted as 1 point. A total score of 0 represented a high level of adherence, scores of 1 and 2 represented medium adherence, and scores of 3 or 4 represented low adherence.

2.3.2. Seizure control

Because decreasing the number of seizures by 50% is the critical point that is often used to distinguish whether a medication is effective, this method was adopted as the measure of seizure control for the current study. The baseline for each patient was considered to be the number of seizures during the preceding six months prior to beginning this program, and this number was self-reported by patients. The number of seizures each patient experienced during the follow-up period was reported according to the patients' records and telephone follow-ups by the pharmacist.

Medication Schedule

This schedule is used to record the antiepileptic drugs and doses that you should take in the morning, at noon, in the afternoon, and at bedtime. To help you to remember to take your medications, please record the most common activities that you are performing at each time period mentioned above.

Medications	Morning☀	Noon☀	Afternoon☀	Bedtime🌙	Others
 Sodium Valproate 500mg/pill	 500mg		 500mg		
 Carbamazepine 200mg/pill	 200mg	 200mg	 200mg		
 Topiramate 100 mg/pill					
 Levetiracetam 500 mg/pill					
Time and things you are doing	8:00 Breakfast	12:00 Lunch	18:00 Supper		

Fig. 1. Medication schedule.

2.3.3. Knowledge of AEDs

A questionnaire was developed to evaluate the level of each patient's knowledge of AEDs, referring to basic items required in the ASHP education guidelines [13] and a questionnaire used in Taiwanese medication education for patients with epilepsy [27]. The five items assessed in the questionnaire were the name of the AED, dosage, duration of AED usage, how to address adverse drug reactions, and missed medications.

2.3.4. Quality of life

The 31-item Quality of Life in Epilepsy Inventory (QOLIE-31) was used to measure the quality of life (QOL) of patients across seven domains: seizure worry, overall QOL, emotional well-being, energy/fatigue, working-driving-social limits, and cognitive function, medication effects. The survey was validated and translated into a number of versions in different languages [28]. The Chinese version was translated and validated by Xiaolin Ren et al. [29].

The seven domains evaluated in the QOLIE-31 were scored using a Likert scoring system with different point values per item. Each subscale and the overall score were calculated and converted to T-scores using a key that can be found in a table in the scoring manuscript of QOLIE-31 [30]. The overall range of T-scores was 11–73.

2.3.5. Other measures

Additional details were measured to elucidate the overall effects of the entire program, such as whether patients forgot to take AEDs, took the missed AEDs, and kept records in a seizure diary at the baseline and at the end of follow-up.

2.4. Study procedure

All patients who met the inclusion criteria were randomly assigned to one of two groups with different levels of intervention. One group was the medication education group (group I), and the second group was the medication education with behavioral intervention group (group II). At the beginning of the implementation, the pharmacist provided all patients in both groups with medication education orally and in writing. Group II was also provided with a modified medication schedule as the behavioral intervention. Then, during the next six months, the two groups were followed up by telephone every month to strengthen the effect of medication education.

Each patient was evaluated and educated by the pharmacist after he/she finished his/her visit. These evaluations occurred immediately following the initial visit in a side room at the clinic either with the

patient alone or with that patient and his/her family. The mean time that the pharmacist spent with each patient was approximately 30 min. The duration of the follow-up phone calls ranged between 5 and 30 min, depending on whether the patient was nonadherent and how many difficulties they had during the follow-up period. A standardized script was used for these aspects of the intervention. In addition, uniform tables were used to ensure that all interventions were performed equally and that the content presented to each patient was the same. The topics that were discussed in the monthly phone calls were items evaluated in this study, and all information was recorded in these tables, such as the time that each patient was contacted via telephone, whether the patient answered the phone, how they

were taking their antiepileptic drugs (AEDs) at the time of the call, whether they missed AEDs and why, and whether they had experienced any adverse effects. The phone calls also provided the opportunity to answer any questions that they had.

Adherence, the knowledge of AEDs, the number of seizures, and other measures were evaluated at the beginning and at the end of the study. The QOL was only measured after intervention. Data were collected by the pharmacist when each patient was recruited into this study, during each monthly follow-up via the telephone, at the patients' visit during the study, and after the end of this program. All measures were compared both between and within the groups. Fig. 2 shows the exact patient flow through the study.

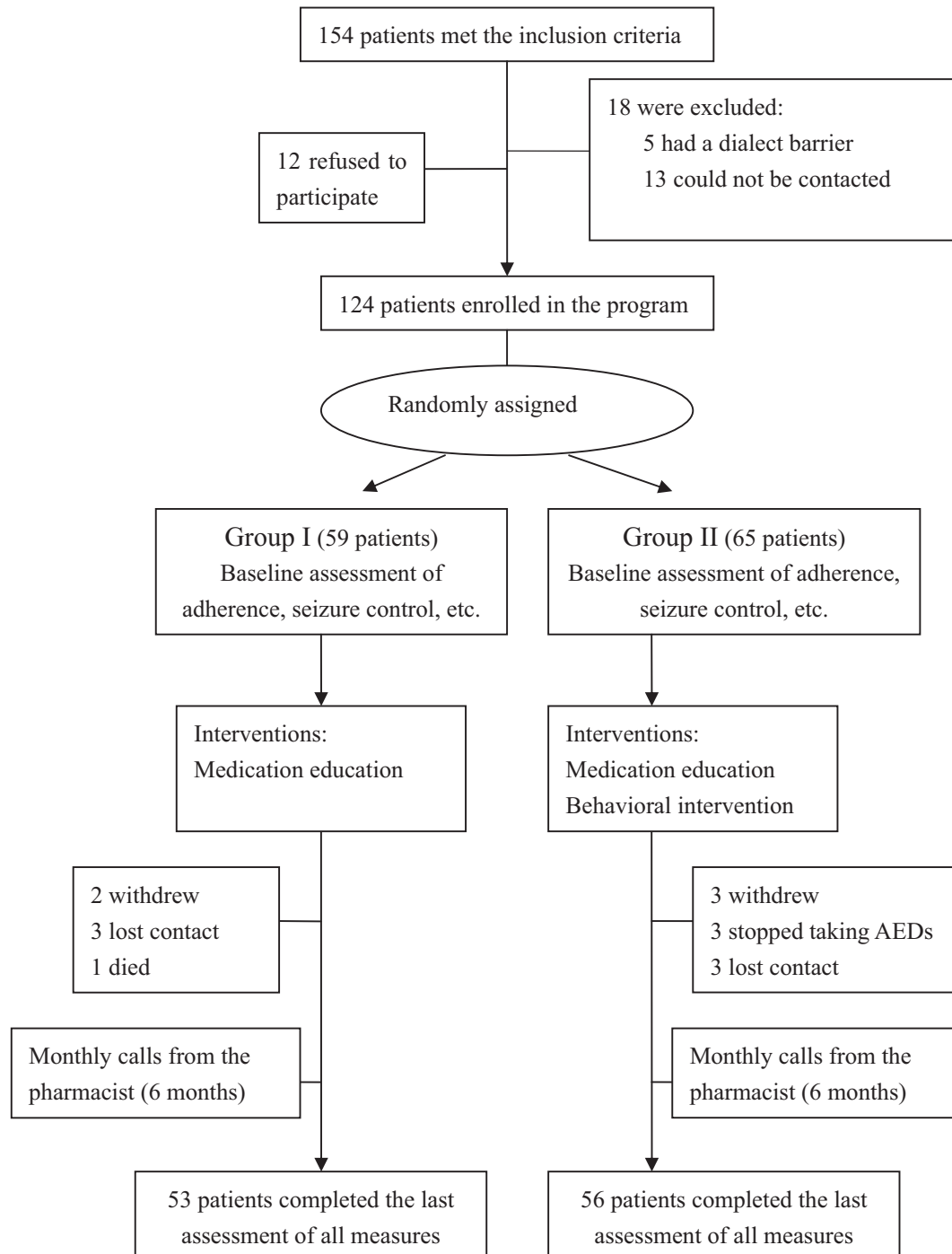


Fig. 2. Patient flow in the study.

Table 1
Baseline characteristics of 109 patients.

Characteristics	Group I (n = 53)	Group II (n = 56)	p-Value
<i>Demography</i>			
Men	26 (49.1%)	33 (58.9%)	0.301
Age, mean (SD), year	31.6 (13.0)	30.8 (11.6)	0.735
Married	27 (50.9%)	31 (55.4%)	0.644
Hired	22 (41.5%)	26 (46.4%)	0.605
<i>Education</i>			
<Primary school	6 (11.3%)	3 (5.4%)	0.630
Middle school	17 (32.1%)	22 (39.3%)	
High school	17 (32.1%)	16 (28.6%)	
College graduate	13 (24.5%)	15 (26.8%)	
<i>Income (per month, ¥)</i>			
<1000	10 (18.9%)	7 (12.5%)	0.534
1000–5000	38 (71.7%)	41 (73.2%)	
>5000	5 (9.4%)	8 (14.3%)	
<i>Disease</i>			
Age of first seizure	20.7 (13.6)	18.9 (11.3)	0.439
Duration of taking AEDs, mean (SD), years	8.1 (7.0)	8.0 (7.3)	0.982
<i>Type of seizure</i>			
Partial seizure	40 (75.5%)	40 (71.4%)	0.633
Generalized seizure	13 (24.5%)	16 (28.6%)	

2.5. Statistical analysis

Descriptive analyses, including the means and standard deviations, were calculated for all quantitative values. Differences between the groups were compared using Pearson's chi-square tests for qualitative values. Student's *t* tests were used to analyze differences in the patients' knowledge and quality of life (overall score). Differences in adherence and the number of seizures following intervention in each group were tested using the Mann–Whitney *U* test. All analyses were performed using the IBM SPSS statistics 19 program.

3. Results

3.1. Baseline characteristics

There were 154 patients who initially met the inclusion criteria, but 12 refused to join, and 18 were excluded. Finally, a total of 124 adult

patients with epilepsy who were non-adherent with their AEDs attended the medication education and behavioral intervention program. Of these 124 patients, 59 patients were randomly assigned to group I, and 65 patients were assigned to group II. During the program, three patients were excluded, five patients withdrew from the program, and seven patients lost contact. Ultimately, 109 patients completed the entire program (Fig. 2). No significant differences in the demographic or disease characteristics of the two groups were observed in the initial 124 patients or in the 15 patients who did not complete the program. The baseline characteristics of the 109 patients included in the final study are listed in Table 1.

3.2. Main outcomes after intervention

No significant baseline differences were observed between the two groups with respect to their adherence ($p = 0.953$), number of seizures ($p = 0.315$), knowledge of AEDs ($p = 0.308$), and history of forgetting to take their AEDs ($p = 0.975$). Similarly, no significant differences were observed between the groups at the end of the follow-up period with respect to their adherence ($p = 0.538$), number of seizures ($p = 0.746$), knowledge of AEDs ($p = 0.537$), quality of life (overall score, $p = 0.947$), and history of forgetting to take their AEDs ($p = 0.946$).

The changes that were observed after medication education and behavioral intervention are provided in Table 2. After intervention, large improvements in knowledge, adherence, number of seizures, and the number of patients forgetting to take their AEDs were observed in group I, as shown in Figs. 3 and 4 and Table 2. Similar outcomes could also be observed in group II, and no significant differences were observed between the groups. However, knowledge increased, adherence improved, and the number of seizures decreased significantly (all of these effects were greater than 60%).

3.3. Details of the knowledge of AEDs and others

Table 3 demonstrates that each item in the knowledge of AEDs improved greatly following intervention, except for knowledge regarding the names of AEDs in group II. This table also indicates that the number of patients who forgot to take AEDs decreased and that the number of recordings in the seizure diary increased following intervention.

Table 2
Main outcomes after intervention.

Difference between baseline and follow-up							
Item	Level	Group I (n = 53)			Group II (n = 56)		
		Baseline	Follow-up	p-Value	Baseline	Follow-up	p-Value
Adherence ^a	Low	12 (22.6%)	0 (0.0%)	<0.000	14 (25.0%)	1 (1.8%)	<0.000
	Middle	37 (69.8%)	21 (39.6%)		37 (66.1%)	18 (32.1%)	
	High	4 (7.5%)	32 (60.4%)		5 (8.9%)	37 (66.1%)	
Seizure times ^a	0	3 (5.7%)	27 (50.9%)	<0.000	1 (1.8%)	22 (39.3%)	<0.000
	1–5	29 (54.7%)	15 (28.3%)		22 (39.3%)	20 (35.7%)	
	6–23	14 (26.4%)	10 (18.9%)		24 (42.9%)	11 (19.6%)	
	24–167	5 (9.4%)	0 (0.0%)		7 (12.5%)	2 (3.6%)	
	≥168	2 (3.8%)	1 (1.9%)		2 (3.6%)	1 (1.8%)	
Knowledge score		1.4 ± 1.0	3.3 ± 1.2	<0.000	1.6 ± 0.8	3.2 ± 1.2	<0.000
Difference between group I and group II							
Outcome		Group I		Group II		p-Value	
Adherence improved		33 (62.3%)		36 (64.3%)		0.827	
Seizure control		34 (64.2%)		36 (64.3%)		0.988	
Knowledge increased		47 (88.7%)		45 (80.4%)		0.231	
Quality of life (overall score)		52.2 ± 9.0		52.4 ± 11.1		0.947	

Knowledge and quality of life (overall score) were analyzed using Student's *t*-test; other factors were measured using Pearson's chi-square test.

^a Analyzed using a Mann–Whitney *U* test.

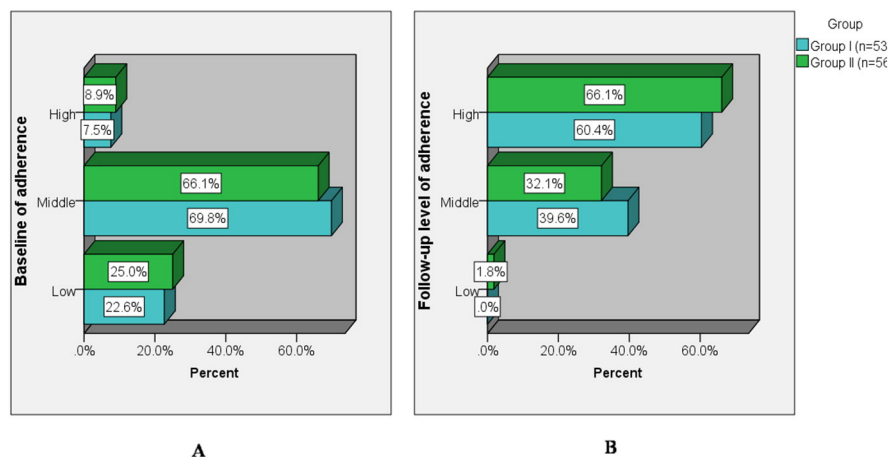


Fig. 3. Adherence before and after intervention.

4. Discussion

To our knowledge, this is the first study examining adherence enhancement in adult patients with epilepsy that compared two different intervention strategies. This study was implemented by a pharmacist in China, and it found no significant effects of combined behavioral intervention on adherence. These results could reflect the fact that the effect of medication education itself was too strong to allow for any additional effect of the modified medication schedule to be observed. However, we did observe that adherence, seizure control, and knowledge of AEDs improved greatly after intervention. Medication education provided by the pharmacist helped to improve adherence, seizure control, and the knowledge of AEDs. Education also helped to increase the number of patients who avoided missing AEDs. Pharmacists can play an important role in the management of patients in the specialized Chinese medical environment. This study demonstrates the role that pharmacists can have in China to improve the utilization of treatment by patients with epilepsy. In future studies, we plan to build upon the information provided here by examining additional measures.

This was the first study to compare the effects of proven interventions for patients with epilepsy in China by a pharmacist. However, a similar trial in rural China [31] demonstrated the practicality and effectiveness of an education package on patients with epilepsy. That study compared education with routine management not with another type of intervention. The 60% reduction in the number of seizures and the improved adherence observed here were greater than the 50% seizure

reduction that was observed in the compliance-enhancing trial in rural China. Based on the outcomes of this program and the trial in rural China, we conclude that AEDs and doctors can play an important role in seizure reduction in rural China.

The decreased number of patients who forgot to take their AEDs and the increased percentage of patients who took their missed AEDs late confirmed the positive effects of medication education in this program. Education has been the most common strategy used by many researchers. Proper education is thought to be fundamental to the successful management of epilepsy and has been found to be effective in enrolling patients into treatment programs, in improving drug adherence, and in markedly reducing nonadherence [1]. However, the effects of education have been moderate [10]. To identify a more effective intervention to improve adherence, we combined education with a behavioral intervention that was intended to help patients incorporate drug administration into their daily lives. This behavioral intervention program was common, inexpensive, and easy to administer. Behavioral interventions may be a helpful tool to increase adherence in patients who regularly forget to take their medications, especially considering the fact that forgetfulness was found to be the primary reason for nonadherence in patients with epilepsy [32]. Disappointingly, the effects of the behavioral intervention program studied here were weaker than expected and lower than the effects of medication education and persistent reinforcement by telephone.

Knowledge of AEDs is poor in China, but medication education provided by pharmacists can help improve this knowledge. A lack of

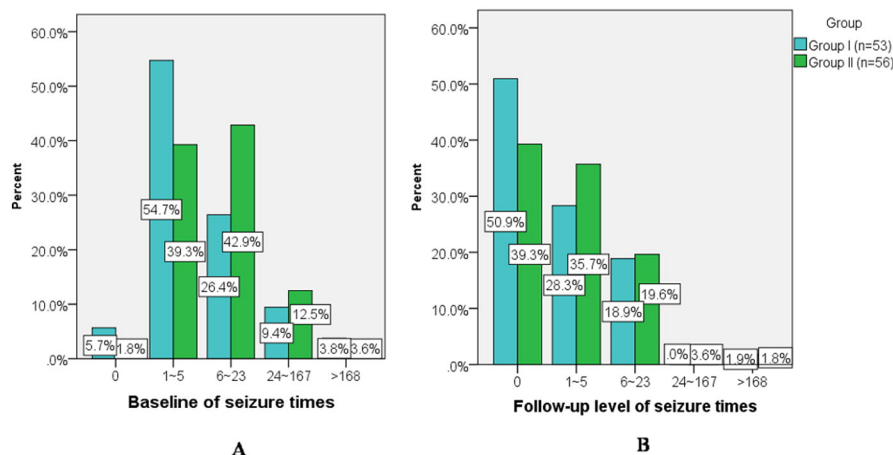


Fig. 4. Seizure frequencies before and after intervention.

Table 3
Details of AED knowledge and others.

Knowledge of AEDs	Group I (n = 53)			Group II (n = 56)		
	Baseline	Follow-up	p-Value	Baseline	Follow-up	p-Value
AED name	44 (83.0%)	51 (96.2%)	0.026	53 (94.6%)	54 (96.4%)	0.647
Dosage	8 (15.1%)	33 (62.3%)	0.000	11 (19.6%)	34 (60.7%)	0.000
Duration	11 (20.8%)	34 (64.2%)	0.000	6 (10.7%)	36 (64.3%)	0.000
Deal with adverse drug reaction	5 (9.4%)	20 (37.7%)	0.001	8 (14.3%)	20 (35.7%)	0.009
Deal with missed medications	7 (13.2%)	38 (71.7%)	0.000	11 (19.6%)	34 (60.7%)	0.000
Forgot to take AEDs	38 (71.7%)	24 (45.3%)	0.006	40 (71.4%)	25 (44.6%)	0.004
Record seizure diary	13 (24.5%)	24 (45.3%)	0.025	10 (17.9%)	28 (50.0%)	0.000

In the 49 (45.0%) patients who forgot to take AEDs, 22 (44.9%) patients took the missed medications later after they remembered (group I: 10 (41.7%) vs group II: 12 (48.0%), $p = 0.656$).

information and/or education regarding epilepsy for patients was demonstrated to be one of the main reasons that epilepsy treatment in developing countries remains far from satisfactory [1]. In addition, the level of patients' knowledge of epilepsy has been shown to differ significantly between countries [33]. Table 3 shows the poor level of the patients' knowledge, in this study. The name of the AEDs was the only piece of information that Chinese patients with epilepsy knew about their medication. The percentage of patients who forgot to take their AEDs decreased to 45.0% from more than 70.0% following their participation in this program, an observation that demonstrates the effectiveness of the pharmacist. The fact that 45.0% of patients forgot to take their AEDs and 44.9% of patients took missed medications told us that forgetfulness remained one of the most important factors in nonadherence. Meanwhile, this study demonstrates that education regarding the importance of not missing medications can decrease nonadherence by half, and knowing how to address missed drugs decreases the number of "forgotten" medications by another half.

Under the Chinese medical system, pharmacists can play an important role in improving the effectiveness of therapy. It is appropriate for pharmacists to take responsibility and focus on patients with special needs, such as nonadherent patients, and the influence of pharmacists may be better and more practical than that of other staff. Clinical visits in China can require patients to travel long distances from various provinces to specific hospitals for the treatment of certain diseases. Patients typically return home after receiving an appropriate therapeutic regimen. They then need to return after several months if the disease recurs or becomes worse. Some patients decide to go to another hospital in search of better therapy. Patients with epilepsy, various dermatologic diseases, and other disorders frequently come to Huashan Hospital. However, the mean time that each patient is able to communicate with doctors in such hospitals in China is only 5–6 min [34,35]. In the current system, doctors typically have to attend to at least 50 patients every 4 h, which results in an average clinical visit duration of approximately 5 min. Because of the limited amount of time that doctors spend with their patients, the level of satisfaction between doctors and patients is low [36], which can be counter-productive [37]. Prior to the intervention, adherence was poor, and patients had frequent seizures. However, after the intervention, adherence for most patients was high, and most patients were seizure-free. These results may largely be attributed to the effects of the physician. Interestingly, 64% appears to be the maximum effect on adherence and seizure control. One factor that might contribute to the low levels of adherence is the short time of the visit. Because 5 min is a limited amount of time, the patients' knowledge of AEDs may not change. However, the patients' knowledge of AEDs increased in this program because of the pharmacist's consistent follow-up every month by telephone. Patients knew the correct expected therapy duration, what type and degree of adverse drug effects were normal, and how to take missed AEDs. Most importantly, as adherence improved, the number of "forgetful patients" decreased. This may indicate that "forgetfulness" can be a type of intentional nonadherence. After the real reasons for forgetfulness are identified, education can lessen patients' nonadherence because patients will not intentionally forget to take their medication. At the same time, this means

that there is a need to distinguish whether patients are actually forgetful. To summarize, there is a great need for pharmacists in China.

Telephone follow-ups may be an important intervention to improve adherence in the unique Chinese medical setting. The unique mode of patient visits in China is a barrier to follow-up appointments. Because many patients live far away from hospitals, they cannot visit hospitals regularly. Telephones are necessary and suitable to facilitate communication with patients and medical staff in China. It is likely that the monthly calls from the pharmacist were principally responsible for most of the observed improvement in patients who only spend 5 to 6 min with doctors, and these calls may be the most important intervention of the pharmacist in this program. Monthly calls from a pharmacist could be considered as a type of specific support regimen, which was found to be critical in medication self-management. This system was originally designed as a way to reinforce the effect of education, but it produced a significant impact on the outcomes. In fact, these calls were ultimately shown to be the most important intervention of the pharmacist. These results suggest a need for further studies to evaluate the effectiveness of this program. Considering the effects of the monthly phone calls from the pharmacist, it is likely that family members who live close to the patients may play an important role in patients' support too. Additional research into the effects of education on families' effect on the patients' regimen-specific support is required, and the role of general social support and other factors remains to be determined.

5. Advantages and limitations

5.1. Advantages

Al-aqeel and Al-sabhan reviewed trials for improving adherence to AED treatment in patients with epilepsy [12]. In that report, the combined patient number in five trials investigating adult patients with epilepsy was 222. In addition, the average follow-up times were generally short, most were only several weeks. Compared with those trials, the number of patients in this trial was moderate, and the follow-up time was longer. In addition to these factors, this trial had several other advantages. One advantage was that many of the study outcomes were evaluated with standard measures, such as the MMAS-4 for adherence assessment and the QOLIE-31 for quality-of-life testing. The MMAS-4 is a validated measure that can be initially used as a diagnostic tool in which patients' adherence behaviors and levels of understanding can be assessed. After specific problems are identified, the appropriate education of the patient can be implemented. For example, patients can be educated to correct their misbeliefs and to help them adapt the patient's regimen to his or her daily schedule to reduce the likelihood of forgetting (e.g., linking medication taking to brushing teeth or eating meals) [19]. This was another reason for choosing the MMAS-4 as the measure to evaluate adherence.

Another advantage of this study was that more details were investigated besides the primary outcomes of adherence and the number of seizures, including the number of patients who forgot to take AEDs and the number of patients that knew how to appropriately address missing a medication dose prior to and following six months

of intervention. Most importantly, this program was shown to be both practical and adaptable in China. This study, which focused on nonadherent patients, was conducted in real clinical settings and performed by a pharmacist when there were too many patients and too little time for doctors to spend an appropriate amount of time with each patient.

5.2. Limitations

First of all, there was no control group that did not receive any interventions or regular follow-ups. Thus, additional research is required to study the impact of these interventions compared with no “treatment” or support. The rationale for not including a control group with no intervention was that the main objective was to determine whether the behavioral strategy was more effective because the effects of education have been proven in many previously published studies. Further, a limited number of patients could be recruited from clinical departments.

Second, we hypothesized that an increased extent of seizure reductions and adherence improvements could be attributed primarily to the doctor in this trial and that the improvements in the other outcome measures resulted from the work of the pharmacist. However, there were no control groups to distinguish between the effects of the doctor and those of the pharmacist. A scale [38] to measure social and regimen-specific support may help determine the impact of this variable. Another shortcoming of the present study was the fact that outcomes were not monitored over a longer time period, such as one year or more.

Third, as there are no records of patient visits across China, we could not evaluate the effects of these interventions on patient visits and cost savings. Therefore, although our results demonstrate that telephone follow-ups with patients reduce the number of times patients came to the hospital, it was difficult to calculate how many hospital visits and how much money was saved.

Finally, the selection of the behavioral intervention lacks theories. Although the feasibility and practicality of the intervention were the major points that were initially considered while selecting the intervention, the behavior of patients was changed by this intervention, including the likelihood that patients would record their seizures in a diary or forget to take their medication. Thus, to achieve improved adherence, we need to design better interventions that are based on behavioral health theories, such as the information–motivation–behavioral skills theory or the health belief model.

Disclosure

The authors declare no conflict of interest. All of the authors confirm that we have read the journal's position on issues involved in ethical publication and affirm that this report is consistent with those guidelines.

Acknowledgements

We thank Professors Kristina M. Cordasco et al. for their permission to modify the medication schedule of the patients in this study. We thank Yi Cai, Zhi-zhong Zhou, Jin Xu and other colleagues in the Department of Pharmacy at Huashan Hospital. We sincerely appreciate their help and preparation for this program.

References

- [1] Sabaté E. Adherence to long-term therapies: evidence for action. World Health Organization; 2003.
- [2] Leppik IE. How to get patients with epilepsy to take their medication. The problem of noncompliance. *Postgrad Med* 1990;88:253–6.
- [3] Faught E, Duh M, Weiner J, Guérin A, Cunningham M. Nonadherence to antiepileptic drugs and increased mortality: findings from the RANSOM study. *Neurology* 2008;71:1572–8.
- [4] Davis KL, Candrilli SD, Edin HM. Prevalence and cost of nonadherence with antiepileptic drugs in an adult managed care population. *Epilepsia* 2008;49:446–54.
- [5] Mac TL, Tran D, Quet F, Odermatt P, Preux P, Tan CT. Epidemiology, aetiology, and clinical management of epilepsy in Asia: a systematic review. *Lancet Neurol* 2007;6:533–43.
- [6] Electroencephalogram And Epilepsy Chapter CSON. Expert consensus opinion in treatment of epilepsy. *Chin J Neurol (China)* 2011;44:56–65.
- [7] Brown I, Sheeran P, Reuber M. Enhancing antiepileptic drug adherence: a randomized controlled trial. *Epilepsy Behav* 2009;16:634–9.
- [8] Cordasco KM, Asch SM, Bell DS, Guterman JJ, Gross-Schulman S, Ramer L, et al. A low-literacy medication education tool for safety-net hospital patients. *Am J Prev Med* 2009;37:S209–16.
- [9] May TW, Pfafflin M. The efficacy of an educational treatment program for patients with epilepsy (MOSES): results of a controlled, randomized study. *Epilepsia* 2002;43:539–49.
- [10] Qureshi NN, Hatcher J, Chaturvedi N, Jafar TH. Effect of general practitioner education on adherence to antihypertensive drugs: cluster randomised controlled trial. *BMJ* 2007;335:1030.
- [11] Roter DL, Hall JA, Merisca R, Nordstrom B, Cretin D, Svarstad B. Effectiveness of interventions to improve patient compliance: a meta-analysis. *Med Care* 1998;36:1138–61.
- [12] Al-aqeel S, Al-sabbah J. Strategies for improving adherence to antiepileptic drug treatment in patients with epilepsy. *Cochrane Database Syst Rev* 2011;19:CD008312.
- [13] Pharmacists ASOH. ASHP guidelines on pharmacist-conducted patient education and counseling. *Am J Health Syst Pharm* 1997;54:431–4.
- [14] Rigsby MO, Rosen MI, Beauvais JE, Cramer JA, Rainey PM, O'Malley SS, et al. Cue-dose training with monetary reinforcement: pilot study of an antiretroviral adherence intervention. *J Gen Intern Med* 2000;15:841–7.
- [15] Kripalani S, Robertson R, Love-Ghaffari MH, Henderson LE, Praska J, Strawder A, et al. Development of an illustrated medication schedule as a low-literacy patient education tool. *Patient Educ Couns* 2007;66:368–77.
- [16] Tordoff J, Simonsen K, Thomson WM, Norris PT. “It's just routine”. A qualitative study of medicine-taking amongst older people in New Zealand. *Pharm World Sci* 2010;32:154–61.
- [17] Sanders MJ, Van Oss T. Using daily routines to promote medication adherence in older adults. *Am J Occup Ther* 2013;67:91–9.
- [18] Paschal A, Hawley S, Romain T, Ablah E. Measures of adherence to epilepsy treatment: review of present practices and recommendations for future directions. *Epilepsia* 2008;49:1115–22.
- [19] Morisky DE, Green LW, Levine DM. Concurrent and predictive validity of a self-reported measure of medication adherence. *Med Care* 1986;24:67–74.
- [20] Zeller A, Schroeder K, Peters TJ. An adherence self-report questionnaire facilitated the differentiation between nonadherence and nonresponse to antihypertensive treatment. *J Clin Epidemiol* 2008;61:282–8.
- [21] Sewitch MJ, Dobkin PL, Bernatsky S, Baron M, Starr M, Cohen M, et al. Medication non-adherence in women with fibromyalgia. *Rheumatology* 2004;43:648–54.
- [22] Hansen RA, Kim MM, Song L, Tu W, Wu J, Murray MD. Comparison of methods to assess medication adherence and classify nonadherence. *Ann Pharmacother* 2009;43:413–22.
- [23] Garbe MC, Nau DP, Erickson SR, Aikens JE, Lawrence JB. The concordance of self-report with other measures of medication adherence: a summary of the literature. *Med Care* 2004;42:649–52.
- [24] Li K, Liang HY, Li X, Chen JP, Zhao L, Yi L, et al. Validity and reliability of the Morisky questionnaire in assessing antihypertensive therapy compliance of hypertension patients in community. *Chin J Hypertens (China)* 2010;18:1067–70.
- [25] Li W, Stewart AL, Stotts NA, Froelicher ES. Cultural factors and medication compliance in Chinese immigrants who are taking antihypertensive medications: instrument development. *J Nurs Meas* 2005;13:231–52.
- [26] Xu WH, Wang Q, Liang WX. Reliability and validity of Morisky questionnaire in measurement of the compliance with hypertensive medications. *Chin J Prev Contr Chron Non-communic Dis (China)* 2007;15:424–6.
- [27] Liu L, Yiu CH, Yen DJ, Chou MH, Lin MF. Medication education for patients with epilepsy in Taiwan. *Seizure* 2003;12:473–7.
- [28] Cramer JA, Perrine K, Devinsky O, Bryant-Comstock L, Meador K, Hermann B. Development and cross-cultural translations of a 31-item quality of life in epilepsy inventory. *Epilepsia* 1998;39:81–8.
- [29] Ren X, Liu X, Liang P. The translation and modified of the quality of life in epilepsy-31 (Chinese version). *Nurs J Chin Pla* 2003;20:99–101.
- [30] Vickrey BG, Perrine KR, Devinsky O, Hays RD, Hermann BP, Cramer JA, Meador KJ, Devinsky O. Quality of life in epilepsy QOLIE-31 (version 1.0): scoring manual. In 1993.
- [31] Li J, Si Y, Hu J, Liu L, Deng Y, He J, et al. Enhancing medical compliance of patients with convulsive epilepsy in rural community: a randomized intervention trial. *Epilepsia* 2013;54:1988–96.
- [32] Tang F, Zhu G, Jiao Z, Ma C, Wang B. Self-reported adherence in patients with epilepsy who missed their medications and reasons for nonadherence in China. *Epilepsy Behav* 2013;27:85–9.
- [33] Doughty J, Baker GA, Jacoby A, Lavaud V. Cross-cultural differences in levels of knowledge about epilepsy. *Epilepsia* 2003;44:115–23.
- [34] Wu D, Lin L, Jiang X. Regularity of outpatients visit time and management strategies. *Chin Nurs Manag* 2007;17:26–8.
- [35] Sun G. Quantitative analysis of visiting time of 14 species of common disease of outpatients in medical institution in Guangzhou. *Med Philos* 2011;32:19–21.
- [36] Zheng X, Ji X. Research on correlations between visit time and patient satisfaction in a variety of investigation ways. *Chin Hosp* 2009;13:39–41.
- [37] Zhang T. Analysis for causes of medical dissensions. *Chin J Hosp Admin* 2005;21:537–9.
- [38] Walker ER, Engelhard George, Christina B, Robin EM, Sterk CE, Dilorio C, et al. A mixed methods analysis of support for self-management behaviors: perspectives of people with epilepsy and their support providers. *Epilepsy Behav* 2014;31:152–9.